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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC-Maths Olympiad at **SUB-JUNIOR LEVEL** *VII to VIII Standards)*

Saturday, 3rd September 2005.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m – 2hours.*

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1. You join a job. Your pay for the first day is Rs. 5/-. Each day after that your pay will be twice as much as it was the day before. Your pay on the tenth will be

   (A) Rs. 100 (B) Rs. 250 (C) Rs. 5120 (D) Rs. 2560

2. Using all the digits 1 to 9 only once, how many nine digit prime numbers can you write?

   (A) 1 (B) None (C) 9 (D) More than 100

3. How many times does the numeral 2 appear in a book having page numbers 1 to 250?

   (A) 81 (B) 106 (C) 55 (D) 76

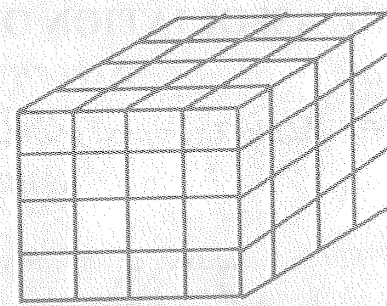
4. In the given addition sum, H, E, O are different digits. The value of the letter O is

$$\begin{array}{r} HE \\ HE \\ HE \\ HE \\ \hline OH \\ \hline \end{array}$$

   (A) 1 (B) 8

   (C) 9 (D) 4

5. A cube of edge 4*cms* is painted externally. It is then cut into one *cm* cubes. How many of these do not have red point on any face?



(A) 4 (B) 8

(C) 56 (D) 16

6. $$\begin{array}{r} A\ B \\ C \\ \hline 4\ 5 \\ \hline \end{array} \qquad \begin{array}{l} A\ B \\ C \\ \hline 9\ 9 \\ \hline \end{array}$$

Look at the additions on the left. Here A, B, C are different digits. If C is placed in the units' column the total is 45. If C is placed in the tens' column the sum is 99. Then the value of A is

(A) 4 (B) 2 (C) 9 (D) 3

7. You have five pieces of 6*cm* rods and 4 pieces of 7*cm* rods. Using some or all of them, which one of the following lengths you cannot measure?

(A) 30 (B) 29 (C) 31 (D) 33

8. Some donkeys and some chickens totally 20 in all were seen in a garden. I counted 64 legs in all. How many donkeys were there?

(A) 12 (B) 10 (C) 14 (D) 17

9. Five boxes and two pencils cost Rs.79. But two boxes and five pencils cost Rs.40. What is the total cost of one book and one pencil?

(A) Rs.39 (B) Rs.19.50 (C) Rs.17 (D) Rs.23.80

10. How many two digit numbers divide 109 with a remainder of 4?

(A) 2 (B) 4 (C) 3 (D) None

11. The highest power of 2 that divides the sum of the numbers

$$4 + 44 + 444 + \ldots + \underbrace{444\ldots\ldots\ldots 4}_{(100 \text{ fours})} \text{ is}$$

(A) 2 (B) 3 (C) 4 (D) 5

12. a and b are square numbers; the l.c.m of a and 140 is 560 and the l.c.m of b and 140 is 700. Then the l.c.m of a and b is

(A) 400 (B) 1600 (C) 2500 (D) 4900

13. a,b,c,d are natural numbers such that $a = bc$, $b = cd$, $c = da$ and $d = ab$. Then $(a+b)\,(b+c)\,(c+d)\,(d+a)$ is equal to

(A) $(a+b+c+d)^2$ **(B)** $(a+b)^2 + (c+d)^2$

(C) $(a+d)^2 + (b+c)^2$ **(D)** $(a+c)^2 + (b+d)^2$

14. If $a^2 + a + 1 = 0$ then $a^2 + \dfrac{1}{a^2}$ is a

(A) positive integer **(B)** positive fraction which is not an integer

(C) negative integer **(D)** negative fraction which is not an integer

15. *ABCD* is a square and an equilateral triangle *CDE* is drawn inward. Then

(A) E lies on AB and $\angle AEB$ is a straight angle
(B) E lies is the interior of the square and $\angle AEB = 150^0$
(C) E lies is the interior of the square and $\angle AEB = 120^0$
(D) E lies outside the square and $\angle AEB = 150^0$

16. Which of the following can never be a common factor of $287 + x$ and $378 + x$ where x can be any natural numbers?

(A) 26 **(B)** 13 **(C)** 91 **(D)** 7

17. How many 4 digit numbers with middle digits 97 are divisible by 45?

(A) 0 **(B)** 2 **(C)** 4 **(D)** 1

18. The product of three consecutive odd numbers is 357 627. What is their sum?

(A) 213 **(B)** 243 **(C)** 153 **(D)** 209

19. The number of integers whose square is a factor of 2000 is

(A) 3 **(B)** 6 **(C)** 10 **(D)** 12

20. A quiz has 20 questions with seven points awarded for each correct answer, two points deducted for each wrong answer and zero for each question omitted. Ram scores 87 points. How many questions did he omit ?

(A) 2 **(B)** 5 **(C)** 7 **(D)** 9